

ProcureGuard Technical & ML Audit Briefing

AI-Assisted Public Procurement Anomaly Engine & Model Evaluation
Theme: SDG 16 — Peace, Justice & Strong Institutions

1. ML Pipeline & 100% Accuracy Audit Analysis

Root Cause of 100% Accuracy in Initial Model:

In synthetic dataset generation, feature distributions between normal and anomalous classes were originally completely disjoint (e.g., `bid_spread_variance` for anomalies was [0.001, 0.015] vs normal [0.04, 0.15]). A single decision tree split achieved zero training error because the classes were unrealistically linearly separable.

Realism Refactoring Applied:

We refactored `dataset_generator.py` to inject realistic stochastic noise, feature overlap (e.g., normal commodity tenders with low bid spread, borderline niche sole-bidders), and 3.5% label ambiguity. The dataset was expanded to **3,500 samples**.

Audited Model Performance:

- **Accuracy:** 96.34% | **ROC-AUC Score:** 0.9314
- **Precision:** 94.94% | **Recall (Sensitivity):** 86.21% | **F1-Score:** 0.9036

2. Trained Feature Engineering Matrix & Gini Importances

Feature Name	Feature Category	Gini Importance	Mathematical Definition & Anomaly Signal
rotation_index	Sequence Pattern	21.7%	Sequence index measuring alternating contract awards among recurring co-bidding vendor groups.
relationship_density	Graph Linkage	20.8%	Density score measuring shared registered address and overlapping corporate directors.
vendor_sole_bid_rate	Competition	19.2%	Proportion of total category awards won by vendor with zero competing bidders.
bid_spread_variance	Bid Spread	13.2%	(Max Bid - Min Bid) / Estimated Value. Low spread indicates cover bidding / artificial price clustering.
ratio_to_estimate	Price Ratio	12.9%	Submitted Bid / Estimated Value. Detects price inflation (>1.15) or predatory under-bidding.
win_rate_share	Market Share	8.5%	Winning vendor's award share in sector (Vendor Wins / Sector Awards).
price_z_score	Category Benchmark	3.6%	(Ratio - Category Mean) / Category StdDev. Category-specific Z-score.
single_bidder_flag	Indicator Flag	0.2%	Binary indicator flag (1 if sole bidder, 0 if multi-bidder tender).

3. Prediction Confidence & UI Integration

- **Investigation Queue Table:** Renders ML Model Confidence % badge inline for every tender in queue.
- **Case Detail View:** Displays AI & ML Anomaly Attribution card with confidence probability gauge and top feature contributions.
- **Settings Lab:** Displays ROC-AUC, Accuracy, Precision, Recall metrics cards and retrain button.